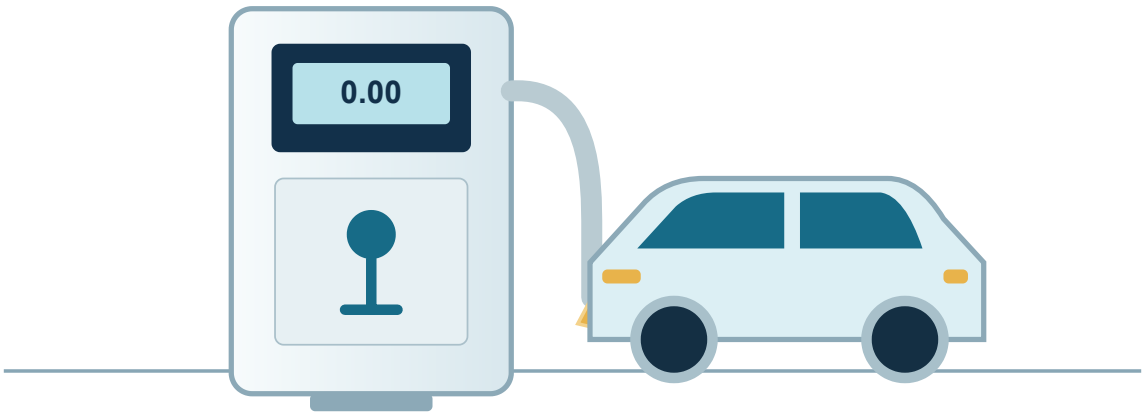
OPERATIONS · SAFE USE · DAILY CHECKS

Fuel Dispenser User Manual

A practical guide to using the fuel dispenser safely and carrying out routine forecourt checks.



FUEL DISPENSER · VEHICLE · SAFE OPERATION

For station operators and customers. Installation, maintenance and repair are covered in separate manuals for authorised technicians.

DOCUMENT GUIDE

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Use this guide to locate operating instructions, safety responses and daily checks.

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Scope note: This manual covers safe use and daily operational checks. Installation, maintenance and repair are described in separate manuals for authorised technicians only.

1–3. Intended Use, Introduction & System Overview

1. Intended Use

Fuel Dispensers are designed to draw fuel in desired quantities from the dispenser tank to the vehicle tank.

This manual provides guidelines to safely use the dispenser and is primarily meant for Station Operators and their customers. It is recommended that people intending to use the equipment, familiarise themselves with the contents of this manual before using dispenser.

Installation, maintenance and repair operations are described in separate manuals for authorised technicians only.

2. Introduction

A fuel dispenser has two major components, one electronic and the other mechanical.

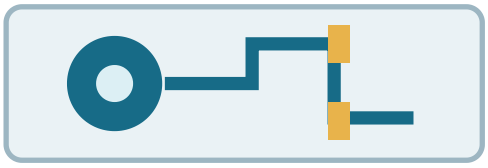
Electronic head

The Electronic head consists of an embedded computer which controls the action of the pump, drives the pump's displays, and communicates with the indoor sales system.



Mechanical section

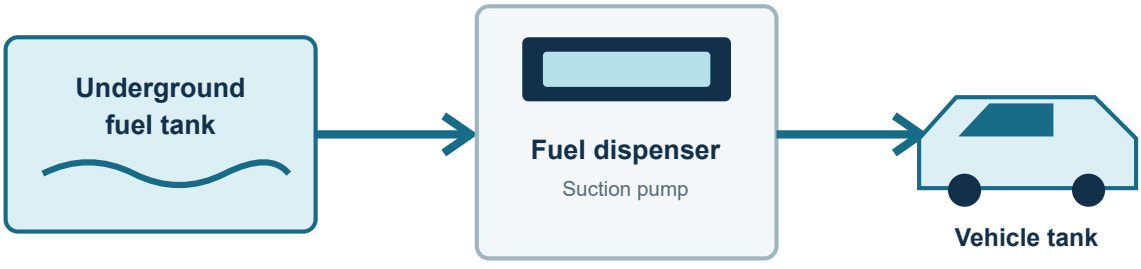
The Mechanical section is a self-contained unit having a motor, pumping unit, meters, pulsers and valves to physically pump and control the fuel flow.



3. System Overview

The fuel to be dispensed is stored in underground tanks. The tanks can hold several thousand litres of fuel. Since the fuel has to be brought up on the ground while the filling it into the vehicle, the pump of the dispenser sucks up the fuel from the underground tank to the dispenser and then transfers the fuel into the tank of the vehicle.

A suction pump transfers the fuel using the principle of unequal pressure.

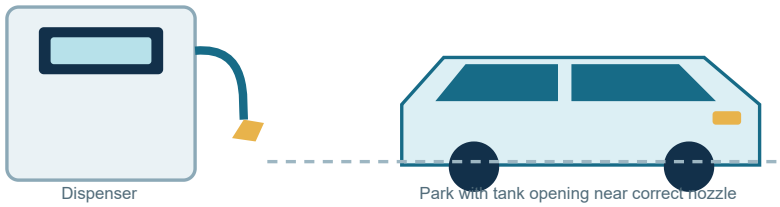


Conceptual flow shown from source description

4. Using the Fuel Dispenser

4.1 Filling Positions

One or more fuel nozzles are provided at each filling position, in the dispenser. Generally, only one nozzle from each filling position may be used at any one time. When a specific nozzle is removed from its holder, the price per litre of the corresponding fuel is displayed on the fuel dispenser.

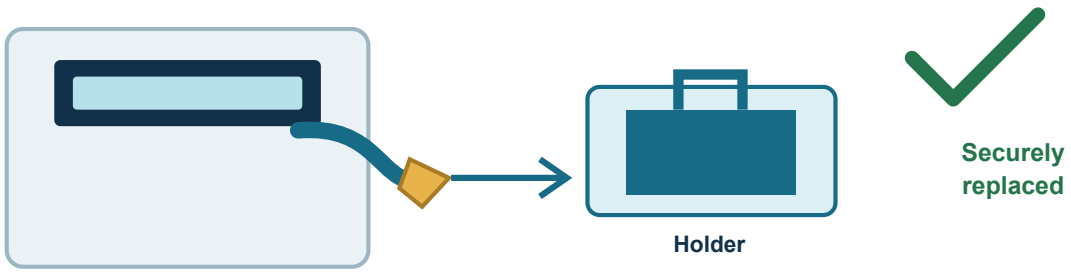


4.2 Steps for Fueling — Steps 1–12

1	Choose a filling position which supplies the correct fuel type; park your vehicle next to the filling position so that the vehicle tank opening is adjacent to the appropriate nozzle. Avoid parking your vehicle in such a way that the nozzle and hose need to be pulled over your vehicle in order to insert the nozzle in the tank opening.
2	Apply the handbrake and switch off the engine of the vehicle. Switch off mobile telephones and extinguish any other sources of fire, such as cigarettes.
3	Remove the lid of your vehicle tank opening. Fuel dispensers may be equipped with a preset function. This allows presetting a monetary or volume value of fuel for purchase. When the preset volume or amount is reached, the fuel dispenser will cease delivery.
4	Select the nozzle which corresponds to the correct type of fuel for your vehicle.
5	Remove the nozzle from its holder. The nozzle is heavy and may not have been securely replaced by the previous user. Take care not to drop the nozzle as it is removed from its holder as this may cause injury or fuel spillage.
6	Verify that the calculator resets to display ‘zero’ for the total volume and amount to pay and, where applicable, the correct unit price for the selected fuel grade. The type of fuel delivered from the selected nozzle is displayed when the nozzle is lifted. If the fuel type displayed is not the same fuel type indicated on the nozzle or above the nozzle holder then replace the nozzle and inform the Station Operator immediately.
7	Do not pull on the hose and do not crush or flatten it. Avoid any contact between the hose and the exhaust pipe under your vehicle. A hot exhaust could burn the hose, causing danger to all people on the forecourt. Take extreme precaution in cold weather as the flexibility of the hose decreases reducing the ease of retraction.
8	Insert the nozzle opening in the tank opening of your vehicle.
9	Pull the nozzle trigger to start the fuel delivery.
10	If the nozzle is fitted with a trigger to hold the nozzle open during delivery, continue to check that the nozzle remains in the tank opening throughout the delivery. **Do NOT leave the nozzle unattended.**
11	Ensure that the hose does not obstruct the alley nor does it constitute a trip hazard.
12	If the nozzle automatically shuts off the fuel delivery, release the nozzle trigger. Wait a few seconds before removing the nozzle from your tank opening to avoid fuel spillage.

WARNING: WHEN REMOVING THE NOZZLE FROM YOUR VEHICLE TANK OPENING, TAKE CARE NOT TO DRIP ANY REMAINING FUEL ON YOUR SHOES, CLOTHES, VEHICLE, OR ON THE FORECOURT. INFORM THE STATION OPERATOR OF ANY SPILLAGES IMMEDIATELY.

4.2 Steps for Fueling — Continued



13 When replacing the nozzle in its holder, take care not to trap your hand or fingers between the nozzle and its holder.

> **WARNING: ENSURE THAT THE NOZZLE IS REPLACED CORRECTLY AND SECURELY IN ITS HOLDER SO THAT IT WILL NOT FALL.**

14 Replace the lid on the vehicle’s fuel tank opening.

15 Ensure that payment is made as required.

16 Drive away from the filling position slowly and carefully.

Before moving away: Confirm the nozzle is returned securely, the tank lid is replaced, and payment is complete. Drive away slowly and carefully.

5. Safety Instructions for the Fuel Attendants

5.1.1 General Safety

- Avoid any spillage of fuel on the forecourt, during vehicle fuel filling and also underground tanks' re-fuelling operations.
- Smoking is prohibited anywhere on the forecourt and also near the fuel dispensers, as indicated by the station's safety instructions.
- Any kind of flames are prohibited on the forecourt.
- Ensure that the engine of the vehicle is switched off, as soon as the vehicle is parked for filling fuel. The engine should only be started, once the filling procedure is over and the customer is ready to leave the fuel station.

5.1.2 Safety in Case of Any Accident

5.1.2.1

Customer Drive Away (with nozzle in vehicle tank)

1. Clean any fuel that has spilled.
2. Check the fuel dispenser for any damages, paying particular attention to the hose and nozzle assemblies.
3. If there is any damage to the fuel dispenser which can compromise safety in use, is apparent or suspected (internally), such as damage to nozzle, swivel, hose, couplings, and so on, then the filling positions involved should be closed and an authorized service person should be called to examine the fuel dispenser for internal and/or external damage. The dispenser should start operation only after any risk of danger from any possible or actual damage has been eliminated.

5.1.2.2

Vehicle Collision on the Forecourt

1. First determine the physical injury, if any that has been caused to people in the involved vehicles or on the forecourt. Take appropriate measures to obtain medical assistance, if required. Ensure that the engines of the involved vehicles are switched off. Ask all people to move away from the area of the collision until any risk of danger is identified and eliminated.
2. Check carefully and individually any relevant fuel dispensers in the collision.
3. Determine the nature and/or risk of damage.
4. The relevant filling positions should be closed and an authorized service person should be called to examine the fuel dispenser for internal and external damage. The dispenser should start operation only after any risk of danger from any possible or actual damage has been eliminated.
5. Clean any fuel that has spilled.

6. Daily Operations

Fuel dispensers with continuous 24 hour operation do not require all opening and closing routines. However the same procedures must be performed at least once a day or at shift change.

6.1 Starting the Service Station

6.1.1 Before

- Visually inspect every fuel dispenser and check for damage of any kind.
- Visually inspect every fuel dispenser to verify that:
  - All warning signs and instructions are present and legible. Replace if required.
  - All emergency equipment is available and in working order.
- Check the base of the dispenser cabinet and the area around all fuel dispensers for any apparent fuel leakage.
- Check all hoses and/or nozzle seals thoroughly.
  - Hoses should be fitted in hose connectors, tightly and all the seals should be intact. Check hoses for any visible signs of leakage, like cracks. Check nozzles or spouts for physical damage (dents, cracks etc.).
- If any damage is evident, contact your service organization for repair or replacement.

6.1.2 During

- Ensure that all nozzles are placed in their holders.
  - As long as the nozzles remain in their holders no fuel delivery will take place.
- Switch on the mains for the pumps of the relevant fuel dispenser(s) at the operating desk.
- If applicable, switch on the fuel dispenser lights.
- Switch on the self-service equipment (if applicable) and put the fuel dispensers into self-service operating mode using the forecourt console.

6.2 During Operation

- In the attended mode, the station attendant should follow the customer instructions as given in the previous sections.
- Check that all lighting areas are working correctly and that the forecourt area is sufficiently illuminated.
  - It is dangerous if customers cannot clearly see what they are doing or are unable to read instructions and/or notices.

6.3 Closing the Service Station

- Check whether anyone is still using the fuel dispensers.
- Be careful not to switch over to standalone mode accidentally. This would allow customers to dispense fuel without restriction.
- Switch off the power supplies to the pumps.
  - Note: Do not switch off any fuel dispensers connected to a self-service system intended for unattended night service.
- Empty waste bins and refill towel and/or plastic glove dispensers as required.

AT-A-GLANCE PROCESS

Dispensing: Quick Reference

Customer fueling sequence summarised from Section 4.2. This page is a quick reference; follow the full instructions and warnings on pages 4–5.



Before dispensing · Steps 1–6

1	Choose a filling position with the correct fuel type and park with the tank opening beside the appropriate nozzle. Do not pull the hose over the vehicle.
2	Apply the handbrake; switch off the engine and mobile telephones, and extinguish other sources of fire.
3	Remove the vehicle tank lid. If using a preset function, note that delivery ceases at the preset volume or amount.
4–6	Select the correct fuel nozzle and remove it carefully. Verify that volume and amount reset to zero and that the unit price is correct where applicable. If the displayed fuel type does not match, replace the nozzle and inform the Station Operator immediately.

During dispensing · Steps 7–12

7–9	Do not pull, crush or flatten the hose, or let it contact the exhaust. Insert the nozzle into the vehicle tank opening and pull the trigger to start delivery.
10–11	Keep the nozzle in the tank opening throughout delivery. **Do NOT leave the nozzle unattended.** Keep the hose clear of the alley and trip hazards.
12	After automatic shut-off, release the trigger and wait a few seconds before removing the nozzle to avoid spillage.
WARNING	When removing the nozzle, avoid dripping remaining fuel on shoes, clothes, the vehicle or forecourt. Inform the Station Operator of any spillage immediately.

Finish safely · Steps 13–16

13	Replace the nozzle without trapping fingers. Ensure it is replaced correctly and securely so it will not fall.
14–16	Replace the vehicle tank lid, ensure payment is made as required, then drive away from the filling position slowly and carefully.

Critical checks: correct fuel type · zeroed display · nozzle attended throughout delivery · nozzle secured in its holder · spills reported immediately.